PHuSE Lab

Dipartimento di Informatica
Università degli Studi di Milano

http://phuselab.di.unimi.it/

CVPL2018, 30-31 AGOSTO, VICO EQUENSE, NAPOLI

PHUSE Lab
Perceptual computing and HUman SEnsing

ACRONYM

Perceptual computing and Human Sensing

Laboratory

HISTORY

LAIV
Analysis of images and
Vision Laboratory

Prof.ssa Paola Campadelli

2015

PHUSE Lab
Perceptual computing
and HUman SEnsing

TEAM

Giuseppe Boccignone
Full professor

Giuliano Grossi
Assistant professor

Raffaella Lanzarotti
Assistant professor

Vittorio Cuculo
Postdoctoral researcher

Alessandro D'Amelio
PhD student

COLLABORATIONS

Università di Torino

SLIPGURU
Università degli Studi di Genova

Laboratoire d'Informatique Fondamentale et Appliquée de Tours

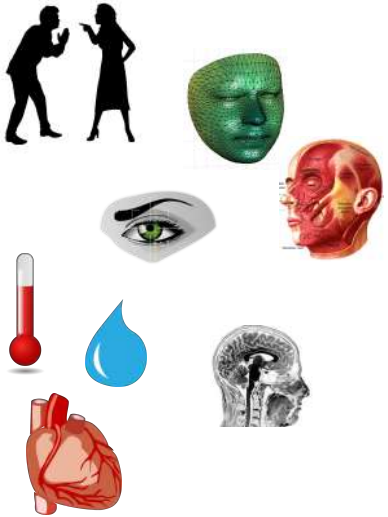
Department of Child and Adolescent Psychiatry

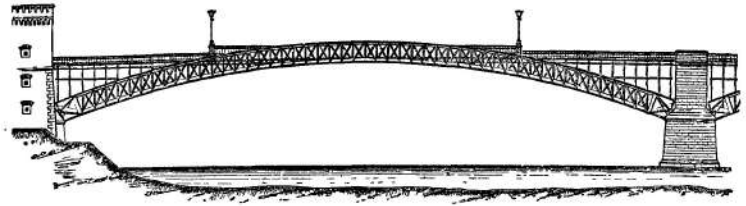
Department of Psychology

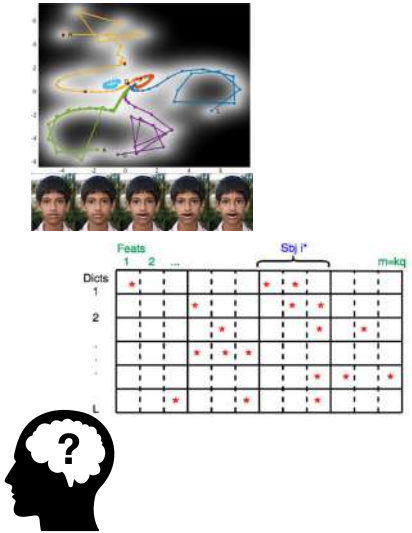
Faculty of Psychology and Laboratory of Action, Perception and Cognition

Experimental Psychology Unit

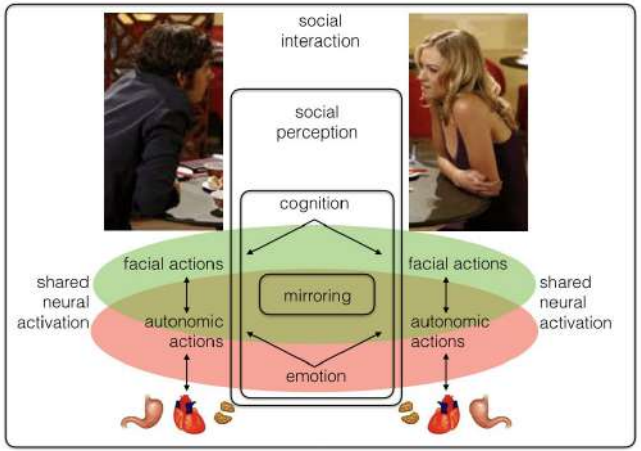
MISSION



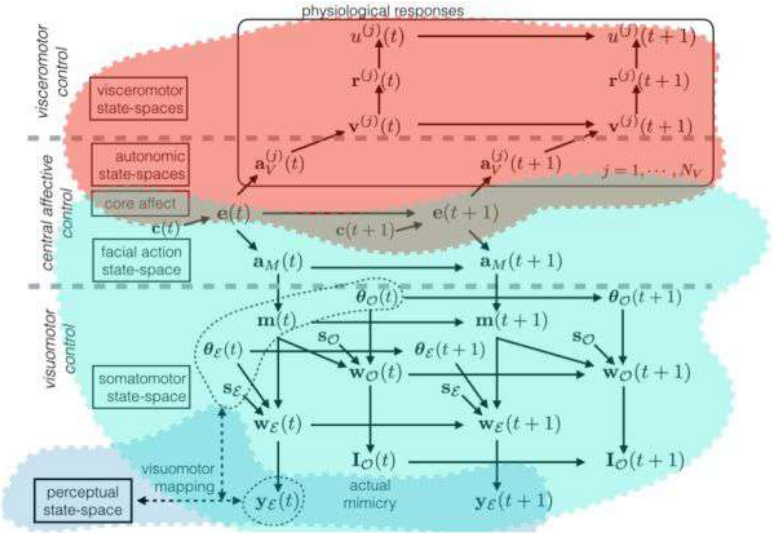




PROBABILISTIC AFFECT MODELLING IN DYADIC INTERACTIONS



a **simulationist approach** to the analysis of facially displayed emotions, in the course of a face-to-face interaction



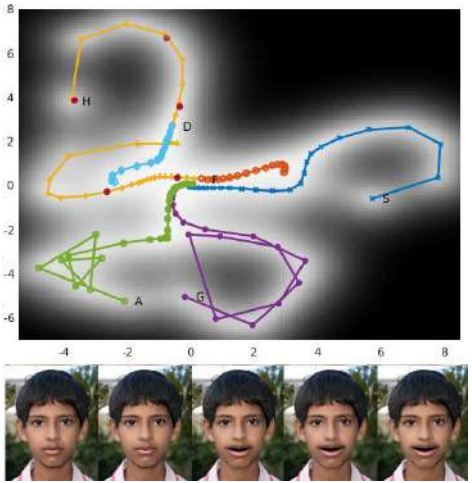
novel probabilistic framework based on a deep latent representation of a **continuous affect space**

Boccignone, G., Conte, D., Cuculo, V., D'Amelio, A., Grossi, G., & Lanzarotti, R. (2018). Deep construction of an affective latent space via multimodal enactment. *IEEE Transactions on Cognitive and Developmental Systems.*

PROCESSING OF FACIAL SIGNALS

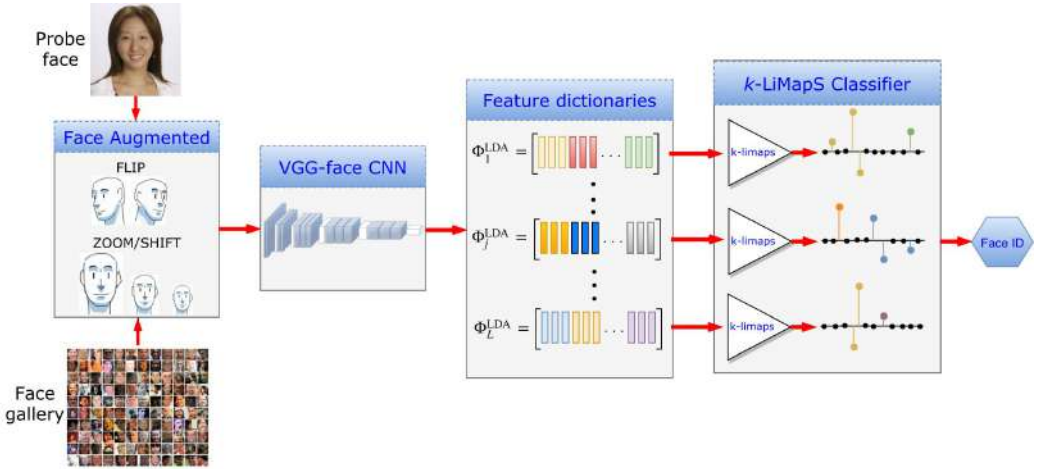
Facial expression

modelling a **somatic motor space** representation of emotional facial expressions



Facial identity

Single Sample Per Person (SSPP) face recognition adopting DCNN + LDA + sparsity based classifier

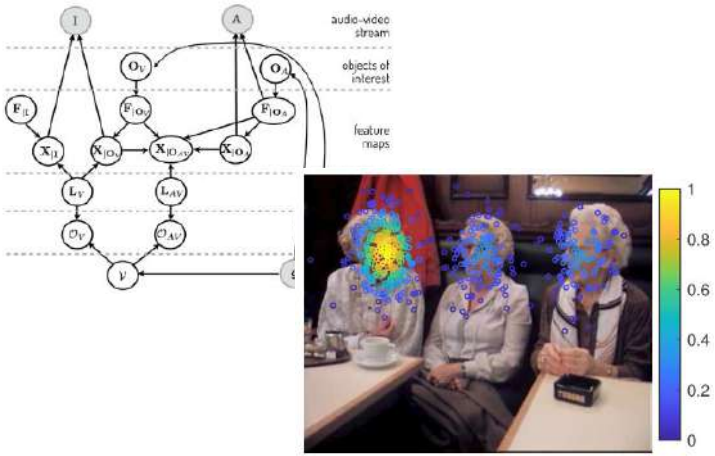


D'Amelio, A., Cuculo, V., Grossi, G., Lanzarotti, R., & Lin, J. (2017, September). A Note on Modelling a Somatic Motor Space for Affective Facial Expressions. In *International Conference on Image Analysis and Processing* (pp. 181-188). Springer, Cham.

Bodini, M., D'Amelio, A., Grossi, G., Lanzarotti, R. & Lin, J. (2018, to appear). Single sample face recognition by sparse recovery of deep-learned lda features. *Advanced Concepts for Intelligent Vision Systems*. Springer,

GAZE PROCESSING

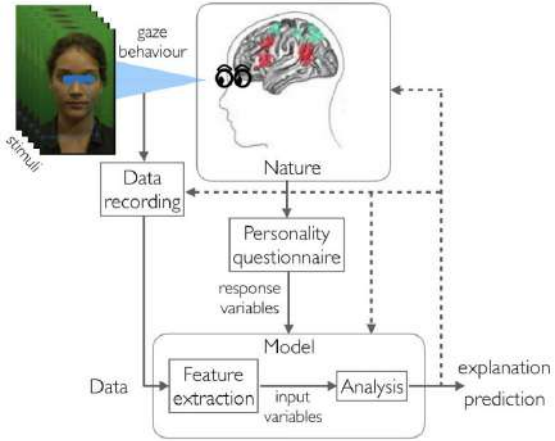
Gaze simulation



perceptual attention to social interactions as displayed in conversational clips, when relying on multimodal information (audio and video).

Gaze analysis

probabilistic classification that relies on Gaussian Processes to find a relation between gaze patterns and **personality traits**

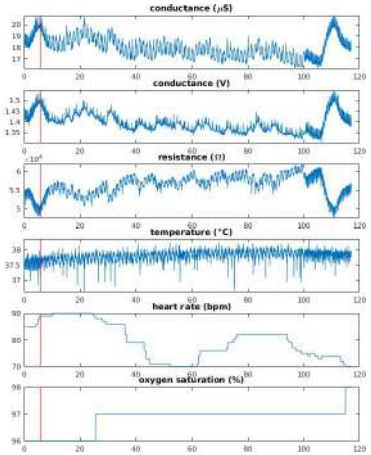


Boccignone, G., Cuculo, V., D'Amelio, A., Grossi, G. & Lanzarotti, R. (2018, accepted). Give ear to my face: modelling multimodal attention to social interactions. In *ECCV'18 Workshop on Human Behavior Understanding*.

Cuculo, V., D'Amelio, A., Lanzarotti, R., Boccignone, G. (2018, to appear). Personality gaze patterns unveiled via automatic relevance determination. In *7th International Symposium From Data to Models and Back (DataMod)*. Springer, Cham.

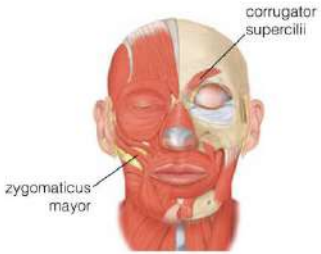
PHYSIOLOGICAL PROCESSING

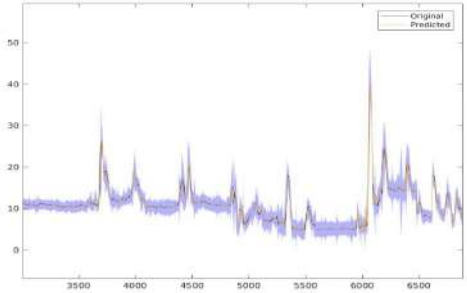
Dataset acquisition



a **novel multimodal dataset** and a **web-based annotation tool** as result of an experiment focusing on amusement emotion

Remote physiological measurement





Boccignone, G., Conte, D., Cuculo, V., & Lanzarotti, R. (2017, November). AMHUSE: a multimodal dataset for HUmour SEnsing. In *Proceedings of the 19th ACM International Conference on Multimodal Interaction* (pp. 438-445). ACM.

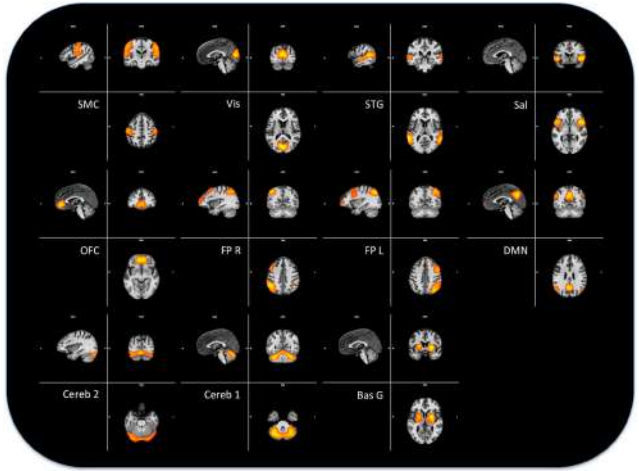
Boccignone, G., Cuculo, V., Grossi, G., Lanzarotti, R., & Migliaccio, R. (2017, September). Virtual emg via facial video analysis. In *International Conference on Image Analysis and Processing* (pp. 197-207). Springer, Cham. **BEST PAPER**

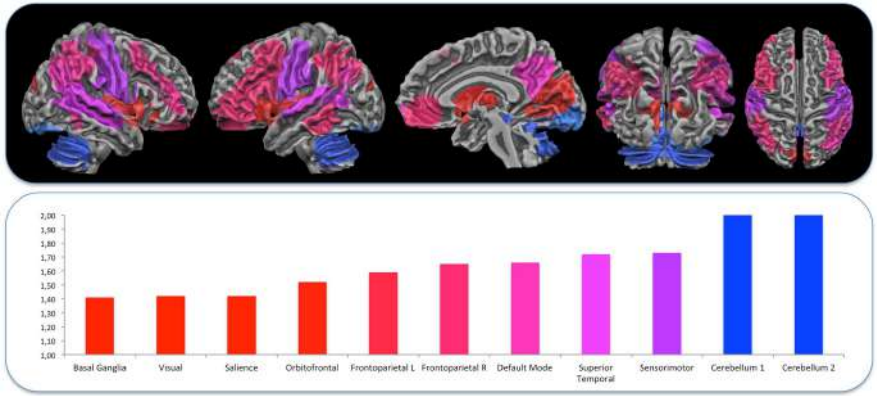
MODELLING FROM fMRI DATA

FOCUSLAB
Functional Neuroimaging and Complex Neural Systems Laboratory

Università di Torino

a series of studies for investigating the relations between the **neural** and **psychological** correlates of **emotions** considering the relations between the **different areas**





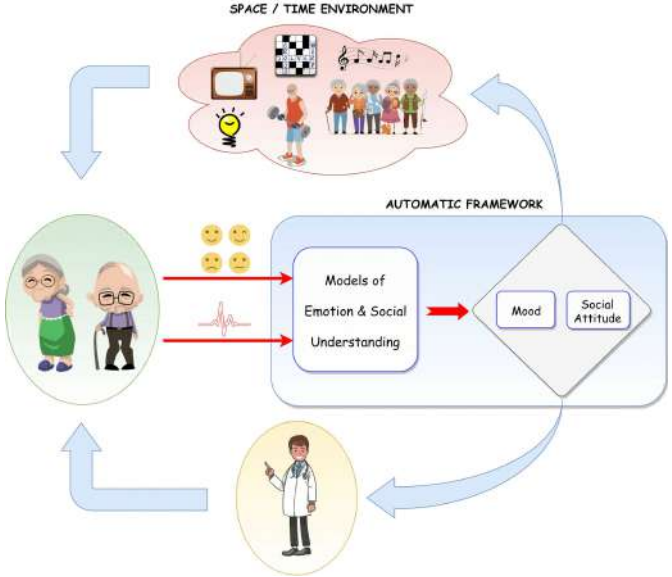
exploiting novel **Bayesian generative modelling** approaches and related computational learning and inference algorithms (e.g. deep learning), we aim to inspect the brain's signals with the intent to infer possible **causal relationships** between different brain areas.

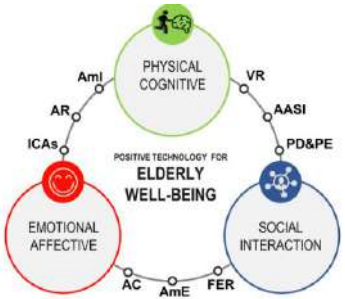
Costa, T., Boccignone, G., Cauda, F., & Ferraro, M. (2016). The foraging brain: evidence of Levy dynamics in brain networks. *PloS one, 11(9).*

USING SOCIAL SIGNALS FOR WELLBEING

SLIPGURU
Università degli Studi di Genova

the **emotional wellbeing** for an aging population is a current issue emerged from the evidence that life expectancy keeps growing, augmenting the proportion of older people over the population





main goal of our research is to design methods for evaluating the emotional wellbeing of **elderly**, including novel computational models and testing procedures to assess the emotional state in an unconstrained, unbiased, and **ecological way**

Grossi, G., Lanzarotti, R., Napoletano, P., Noceti, N., Odone, F. (2018, submitted). Intelligent Systems for Physical, Cognitive, Emotional and Social Well-being of Older People: A Review. *Pattern Recognition Letters*.

Thanks

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